AP PRECALCULUS

Unit 1 • Lesson 1.3 • Guided notes

Topic 1.3 • Rates of Change in Linear and Quadratic Functions

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Recognize linear functions by their constant rate of change.
- Recognize quadratic functions by their constant second differences, and find the quadratic from a table.
- Connect the sign of the leading coefficient to concavity and to the vertex.

Vocabulary. linear function, slope, quadratic function, first differences, second differences, vertex, axis of symmetry.

1. Linear functions

Key idea

A linear function $f(x) = mx + b$ has the _____ average rate of change, equal to _____, over every interval. In a table with equally spaced inputs, the outputs change by a _____.

2. Quadratic functions

Key idea

For a quadratic $f(x) = ax^2 + bx + c$, the average rates of change over consecutive equal-length intervals change at a _____ rate. With equally spaced inputs, the _____ are constant. When the input step is 1, the second differences equal _____.

Example 1 • Classify each table

x	0	1	2	3
A	4	7	10	13
B	1	2	5	10
C	1	2	4	8

Example 2 • Finding the quadratic

x	0	1	2	3	4
y	3	1	-3	-9	-17

Show that the data are quadratic and find the function.

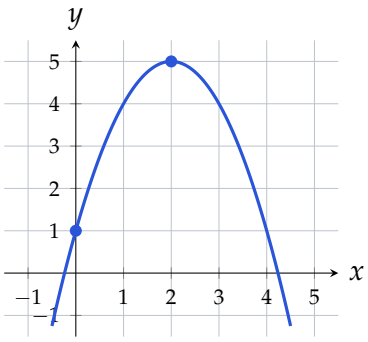
3. Concavity and the vertex

Key idea

If $a > 0$, the rate of change is _____: the parabola is concave _____ and has a _____ at the vertex.

If $a < 0$, it is concave _____ with a _____. The vertex is at $x =$ _____; there the rate of change changes sign.

Example 3 • From a graph



Write an equation for this parabola and compute the average rates of change on $[0,1]$, $[1,2]$, $[2,3]$.

Common mistakes

- Using differences when the inputs are not equally spaced.
- Thinking the second difference equals a; it equals $2a$ when the step is 1.

Check your understanding

1. Outputs $2, 5, 10, 17$ at $x = 0, 1, 2, 3$ come from a _____ function.
2. Second differences of 6 (step 1) mean $a =$ ____, and the graph is concave _____.